

CARLOS JAVIER TANGARIFE GIL

Senior Software Engineer | Tech Lead | Full Stack Developer

📍 Medellín, Colombia | 📞 +57 3012295992
🔗 [Linkedin](#) | 🌐 Available for Remote Work
🔗 <https://carlostangarife.com/>

EXECUTIVE SUMMARY

11+ years of proven excellence in full-stack software engineering, with deep expertise in Cloud Architecture (AWS, Azure), Frontend (React, Angular), Backend (.NET, Node.js), and DevOps/CI-CD. Successfully reduced technical debt by 30-70% across multiple enterprise projects while leading digital transformations for US-based companies with 3M+ active users.

Key Differentiators:

- 🏆 Reduced 200 AWS Lambda functions to 60 (~70% reduction) while improving performance
- 🏆 Reduced 160 .NET legacy projects to 85 (~47% reduction) through strategic refactoring
- 🏆 Transformed 2-day database operations to 2 hours through query optimization
- 🏆 Achieved weeks-to-minutes deployment cycles through CloudFront automation
- 🚀 AI-Assisted Development advocate using Cursor, ChatGPT, and automated workflows

CORE COMPETENCIES

Frontend Engineering React • Angular • Vue • TypeScript • JavaScript • Vite • Webpack • Chakra UI • Ionic • Web Components • Storybook • SASS/SCSS • HTML/CSS • Responsive Design • Mobile-First	Backend Engineering Node.js • NestJS • Express • C# • .NET Core • .NET Framework • Clean Architecture • Hexagonal Architecture • SOLID • CQRS • Microservices • Monoliths • MVC • Dependency Injection • SignalR • API Design
Cloud & Infrastructure — AWS Lambda • API Gateway • DynamoDB • S3 • SNS • SQS • Cognito • CloudWatch • CloudFront • Secrets Manager • Parameter Store • CodePipeline • CodeBuild • CodeDeploy	Cloud & Infrastructure — Azure App Service • DevOps • Key Vault • Blob Storage • SQL Database • Redis Cache • Service Bus • Container Registry • Application Insights • AD B2C
Database & Data PostgreSQL • SQL Server • MongoDB • Oracle • TypeORM • LINQ • Query Optimization • Indexing • Materialized Views • PL/SQL	Testing & Quality Vitest • Cypress • Playwright • Selenium • SpecFlow • Jest • Jasmine • Unit Testing • Integration Testing • E2E Testing • BDD/TDD • Gherkin • SonarQube • OWASP
DevOps & Tools Docker • CI/CD Pipelines • Azure DevOps • GitHub • GitLab • Jenkins • Serverless Framework • Hangfire • Git • Jira • Trello	Methodologies & Practices Agile/Scrum • Clean Code • DDD • Pattern Design • Code Review • Pair Programming • Documentation (Notion) • Feature Flags (ConfigCat) • Semantic Versioning

PROFESSIONAL EXPERIENCE

SOMBRA Senior Software Engineer (Remote) Client: CampMinder (USA) — Camp Management Software Platform 📅 November 2024 – Present 📍 Remote
--

Leading modernization initiatives for a legacy camp management platform serving diverse subscriptions with complex role-based permissions and feature flag configurations across multiple client contracts.

Frontend Development & Architecture

Tech Stack: React • TypeScript • Vite • Chakra UI • React Query • React Hook Form • Zod

- Architecting and implementing mobile-first responsive UI for legacy application modernization
- Building complex menu systems filtered by granular permissions and feature flags (ConfigCat)
- Designing advanced data caching strategies with React Query: custom cache policies, staleTime, refetchOnWindowFocus, pagination, optimistic updates
- Creating reusable component library with React Context for state management across application

Communication Hub (Comms Hub) — Core Team Member

- Developing enterprise CMS platform for multi-channel messaging (SMS, Email, WhatsApp) via Twilio
- Building customizable email templates using BeePlugin for drag-and-drop editing
- Architecting Monorepo structure (Turborepo) with shared component packages across multiple applications
- Implementing semantic versioning and publishing internal packages to Azure CDN (primary) and NPM (fallback)
- Ensuring high availability with Azure CDN optimized for Colorado, US regions

DevOps & CI/CD Leadership

- Designing and maintaining YAML pipelines across multiple environments in Azure DevOps
- Integrating Docker, Azure App Service, Key Vault, and quality gates
- Implementing semantic versioning for package releases with automated changelog generation
- Publishing and distributing internal packages outside monorepo for company-wide reuse
- Coordinating with DevOps team for elevated permissions (Secrets, Key Vaults)

AI-Driven Development Innovation

- Championing AI adoption initiative using Cursor (AI-integrated editor) and ChatGPT
- Developed automated branch creation workflows analyzing user story titles/contexts with AI
- Using AI to infer optimal branch naming and task segmentation strategies
- Defining precise user stories and acceptance criteria using Gherkin Language
- Curating high-quality prompts while maintaining human supervision for production safety
- **Result:** Increased delivery speed and consistency without compromising code quality

Backend Modernization & Legacy Refactoring

- Working across mixed .NET environment (.NET Framework 4.x to .NET 8)
- Reduced 160 legacy projects to 85 (~47% reduction) through systematic refactoring and de-duplication
- Migrating monolithic classes toward Clean Architecture, SOLID, and dependency injection
- Optimizing SQL Server queries: indexing strategies, normalization, complex joins for performance
- Refactoring large stored procedures and improving database schema design

Quality Assurance & Observability

- Implementing automated testing with Vitest (unit) and Cypress (E2E)
- Integrating Bugsnag for silent error tracking in production with automatic ticket creation
- Telemetry-driven prioritization: automatic Azure DevOps work items from production errors
- Building comprehensive Notion documentation for technical specs, workflows, and business processes

Agile & Collaboration

- Operating in mature Scrum environment with Azure Boards for backlog management
- Pair programming sessions with tech leads for knowledge transfer
- Leading collaborative refinement sessions bridging business and engineering teams

Key Achievements

- **✓ Reduced technical debt by ~30% through systematic refactoring and clean architecture adoption**
- **✓ Improved CI/CD reliability through ConfigCat integration and custom Azure CDN distribution strategy**
- **✓ Accelerated delivery cycles by 40% through AI-assisted coding and workflow automation**
- **✓ Strengthened documentation culture ensuring traceability between business requirements and technical implementation**
- **✓ Established internal package ecosystem enabling code reuse across 10+ applications**

Technologies: React, TypeScript, Vite, Chakra UI, React Query, React Hook Form, Zod, ConfigCat, Twilio, BeePlugin, Turborepo, Azure DevOps, Docker, Azure App Service, Azure Key Vault, Azure CDN, NPM, Cursor, ChatGPT, Gherkin, Notion, Vitest, Cypress, Bugsnag, .NET

AVENUE CODE | Senior Full Stack Engineer (Remote)

Client: Blink Fitness (USA) — Nationwide Gym Chain Platform | 🌐 2022 – November 2024 | 📍 Remote

Developed and optimized mission-critical mobile and web applications for nationwide gym chain serving 3+ million active users, managing memberships, personal training, nutrition programs, and real-time facility tracking.

End-to-End Feature Development

Built comprehensive cross-platform solution (web + mobile) featuring:

- **Nutrition & Wellness:** Personalized content delivery based on membership tier
- **Training Management:** Scheduling, rescheduling, canceling personal training sessions with real-time availability
- **Goal Tracking:** Progress monitoring for fitness objectives with visual dashboards
- **Facility Check-in/Check-out:** Membership validation, retention flows, and access control
- **Crowd Tracker:** Real-time gym occupancy monitoring helping users avoid peak hours and optimize visit timing

Frontend Development

- Built high-performance UIs with Angular, Ionic, TypeScript, Capacitor, and SASS
- Leveraged Ionic's native components for consistent iOS and Android experiences
- Implemented membership-based access control with tiered features (Green, Silver, Basic)
- Differentiated features by tier: guest privileges, multi-location access, premium training content
- Optimized rendering performance for mobile devices with limited resources

Dynamic Content Management System

- Architected dual-CMS platform: WordPress (backend content) + Contentful (frontend config)
- Achieved full dynamic control: text, routing, colors, icons, UI states without code deployment
- Repurposed Contentful as feature-flag system enabling/disabling modules by membership tier
- Implemented real-time content updates without app store submissions

Serverless Backend Architecture

- Architected fully serverless backend with AWS Lambda, DynamoDB, S3, API Gateway
- Implemented secure authentication via AWS Cognito with MFA support
- Built centralized monitoring using CloudWatch and Datadog with domain/client filtering
- Refactored 200 Lambda functions to 60 (~70% reduction) eliminating deprecated code
- Applied Clean Architecture and SOLID principles for maintainability and scalability
- Decoupled business logic, data access, and third-party integrations for testability

Event-Driven Architecture

- Implemented asynchronous communication using AWS SNS and SQS
- Built event-driven workflows enabling resilient, scalable inter-service communication
- Designed retry mechanisms and dead-letter queues for fault tolerance

Automated Email Notification System

- Developed dynamic email system retrieving HTML templates from S3 buckets
- Implemented template population engine with customer-specific data
- Delivered personalized emails for: membership updates, class schedules, promotional campaigns
- Integrated with membership lifecycle events for automated triggered emails

Infrastructure as Code (IaC)

- Adopted Serverless Framework for infrastructure automation
- Single-command deployments for Lambda functions, DynamoDB tables, API Gateway configs
- Environment-specific configurations with parameter management

Security & Configuration Management

- Centralized secrets using AWS Systems Manager Parameter Store and Secrets Manager
- Implemented hierarchical access control for credentials and configuration values
- Automated secret rotation and compliance auditing

CI/CD Innovation

- Built automated pipelines with AWS CodePipeline, CodeBuild, CodeDeploy
- **Revolutionary Hot Deployment:** Transformed semi-manual App Store/Play Store submissions into instant CDN updates
- Converted Ionic container into CloudFront CDN-rendered asset delivery model
- Eliminated store submission dependency for non-native updates
- Reduced release cycles from weeks to minutes with full automation
- Achieved cost savings vs third-party solutions (Ionic AppFlow)

Key Achievements

-  **Refactored 200+ Lambda functions to 60 (~70% reduction), dramatically reducing maintenance overhead**
-  **Automated mobile deployment pipeline through CloudFront, cutting release time by 95%**
-  **Introduced Clean Architecture for AWS serverless systems, enabling team scalability**
-  **Reduced infrastructure costs by 60% migrating from third-party tools to in-house AWS solution**
-  **Improved system observability with integrated Datadog and CloudWatch pipelines**
-  **Supported 3M+ active users with 99.9% uptime during peak hours**

Technologies: AWS (Lambda, API Gateway, DynamoDB, S3, SNS, SQS, Cognito, CloudWatch, CloudFront, Secrets Manager, Parameter Store, CodePipeline, CodeBuild, CodeDeploy), Angular, Ionic, TypeScript, Capacitor, SASS, WordPress, Contentful, Serverless Framework, Datadog, Clean Architecture, SOLID, Scrum

DITECH | Senior .NET Consultant (Remote)

Client: Grupo Sesé (Spain) — European Logistics Provider | 📅 2021 | 📍 Remote

Digitalized logistics workflows for leading European logistics provider across warehouses and retail hubs throughout Europe, managing inbound/outbound flows, inventory, packaging, and shipment tracking.

Backend Development

- Built .NET Core (C#) solution following MVC pattern with Dependency Injection
- Implemented customer-specific monolithic instances with shared codebase for tailored customizations
- Designed SQL Server transactional data layer with LINQ for type-safe querying
- Implemented Hangfire for background job scheduling: inventory reconciliation, shipment reports, nightly updates
- Ensured consistency across distributed warehouses with automated synchronization

Frontend Development

- Delivered dynamic interfaces with Kendo UI for warehouse personnel
- Built responsive, data-rich views with AJAX for asynchronous operations
- Optimized data refresh cycles for users handling thousands of items daily
- Implemented complex grid operations: filtering, sorting, grouping, inline editing

Quality Assurance & BDD

- Followed Behavior-Driven Development (BDD) with stories in Gherkin
- Trained team on writing clear, testable user stories
- Built automated testing pipelines with SpecFlow, Selenium, and Playwright
- Integrated CI/CD through Azure DevOps for continuous delivery confidence

Case Study: Internal Employee Engagement Platform

Challenge: Internal innovation project for employee recognition using Ethereum smart contracts for digital "coins" suffered severe performance issues: slow load times, high latency, unresponsive UI from improper MVC Razor view rendering blocking entire page generation.

Solution: Segmented heavy queries into smaller async calls; rebuilt monolithic views into partial views loaded dynamically with Kendo UI AJAX; parallelized backend service requests for partial rendering.

-  **Reduced page load time by 90%, transforming user experience**
-  **Restored user trust, engagement soared after relaunch**
-  **Successfully enabled Ethereum-based token transactions for employee recognition**
-  **Turned neglected initiative into recognized internal success story**

Technologies: .NET Core, C#, MVC, Dependency Injection, Kendo UI, Monoliths, LINQ, Unit Testing, Integration Testing, Mocks, Gherkin, Selenium, Playwright, SpecFlow, SQL Server, Azure DevOps, CI/CD, HTML, CSS, JavaScript, Scrum, Hangfire, Ethereum Smart Contracts

CEIBA SOFTWARE | Developer and Architect

📅 2019 – 2021 | 📍 Medellín, Colombia (On-site)

Evolved into client-facing technical leader for large-scale enterprise projects, aligning technical solutions with business needs. Became internal coach for Angular/TypeScript, promoting enterprise architectural principles (DI, IoC, SOLID) across front-end teams.

Leadership & Mentorship

- Organized workshops and internal talks on Angular and TypeScript best practices
- Mentored developers on scalable front-end patterns and reusable component design
- Acted as technical coach advising teams on clean architecture principles
- Collaborated directly with stakeholders to understand challenges before development
- Helped clients evaluate third-party services vs custom solutions, building long-term trust

PROJECT: VENNDELO (2020 – 2021) — Web marketplace for product sales and supplier distribution

Technical Challenges

- Replace legacy PHP systems with modern architecture
- High flexibility for diverse product catalogs, dynamic pricing, supplier relationships
- Seamless integration of Vue.js components into legacy AngularJS system

Key Contributions

- Technical Advisor & Developer guiding team through Gherkin syntax for clearer business logic
- Established shared understanding between developers, QA, and business stakeholders
- Implemented automated E2E testing with Twine and Gherkin for critical path verification

Webpack Mastery

- Mastered Webpack at native level for complex build configuration
- Combined legacy AngularJS with new Vue.js frontend built entirely with Web Components
- Restructured Webpack to compile components dynamically at build time
- Every UI element packaged as reusable Web Component for legacy embedding
- Drastically improved performance and made integration pipeline stable across environments

Backend Architecture

- Established NestJS backend inspired by Angular's modular structure
- Introduced Clean Architecture principles separating concerns for testable business logic
- Leveraged TypeORM with PostgreSQL for scalable persistence layer

Technologies: Webpack, Storybook, Element UI, Foundation, SCSS, TypeScript, Node.js, NestJS, Ionic, Web Components, HTML, JavaScript, PHP 7, PostgreSQL, TypeORM, Angular

PROJECT: RENTING COLOMBIA (2019 – 2020) — Leading vehicle leasing and mobility provider

The Challenge: Ecosystem of 15 interconnected Angular applications developed by different teams with different Angular versions and inconsistent practices. Every app duplicated shared components creating bugs, performance issues, and unsustainable maintenance.

Front-End Rearchitecture Leadership

- Led standardization and unification effort across entire platform
- Designed shared Angular library with Angular Material centralizing rendering logic, permissions, and layouts
- Package became core foundation reducing redundancy and ensuring consistent UX
- Defined best practices for dependency injection, module creation, enterprise structure using strongly-typed TypeScript

Performance Optimization

- Problem: Pages rendering up to 20,000 elements at once causing severe bottlenecks
- Solution: Implemented infinite scrolling and lazy loading
- Coordinated with backend teams to enforce paginated API responses across all services
- Optimized change detection strategies and OnPush components

Security Enhancement

- Redesigned cross-application authentication around Azure AD B2C and MSAL
- Built centralized authentication flow: token validation, permission exceptions, password management
- Created unified login experience across all 15 applications

Key Achievements

-  Transformed fragmented suite into cohesive, modular, high-performing platform
-  Became reference architecture within Ceiba for multi-app ecosystems
-  Reduced development time for new features by 50% through shared component library
-  Improved performance by 80% through optimization strategies

Technologies: .NET Core, C#, Hexagonal Architecture, MVC, Microservices, Pattern Design, Dependency Injection, SignalR, CQRS, Azure Redis Cache, Azure Service Bus, Azure Table Storage, Azure Container Registry, Azure App Service, Azure Blob Storage, Azure SQL Database, Azure Key Vault, Azure Application Insights, Azure Active Directory B2C, Docker, CI/CD, Azure DevOps, LINQ, Angular

XPERTGROUP | Developer

 2018 – 2019 |  Medellín, Colombia (On-site)

Custom development, analytical solutions, machine learning, and business intelligence services.

PROJECT: CFA — Credit Scoring Platform (2018 – 2019)

Dynamic credit scoring platform with runtime-configurable variables and formulas.

Process Transformation

- **Before:** Legacy documentation-driven process with long requirement documents causing slow iteration
- **Action:** Proactively promoted Agile and Scrum practices
- **Result:** Introduced incremental development cycles with real-time client review, refinement, and approval

Technical Implementation

- Designed highly configurable scoring engine allowing stakeholders to define variables and formulas at runtime without developer intervention
- Built flexible data model and dynamic form rendering system in Angular and TypeScript
- Implemented on-the-fly field creation: dropdowns, checkboxes, text fields, numeric ranges
- Each field type with customizable validation rules and error handling
- Integrated EasyChart for interactive score visualization

Business Impact

-  Empowered business users to modify credit evaluation logic directly through UI
-  Eliminated technical dependency for every rule change
-  Achieved minimal delay between policy changes and production rollout
-  Created self-service platform mirroring real-world financial decision-making

DevOps & Quality Automation Initiative

- Led creation of DevOps automation framework from ground up when CI/CD practices were emerging
- Implemented Jenkins for automated build and deployment pipelines
- Integrated SonarQube for continuous code quality analysis
- Tracked: code coverage, executed tests, success/failure rates, rejected PRs, code duplication

Key Achievements

-  Established CI/CD culture before it became standard practice in company
-  Achieved 95%+ code coverage maintained through automated gates
-  Reduced bug rate by 70% through continuous quality analysis
-  Client satisfaction at launch: fully stable, measurable, compliant system

Technologies: C#, .NET Core, Angular, React, OOP, DevOps, Jenkins, Unit Testing, Integration Testing, SonarQube

NEWSHORE | Advanced Developer

 2017 – 2018 |  Manizales, Colombia (On-site)

Digital solutions for airlines worldwide. First software company role establishing technical foundation. Worked with European low-cost airlines: Vueling, Volotea.

PROJECT: VUELING — Flight Booking Platform (2017 – 2018)

Technical Implementation

- Built configurable search engine: origins, destinations, passenger types, promotional periods
- Made system fully dynamic and reusable for adaptation to other airlines with minimal code changes
- Platform: C#, .NET Framework, Angular, Umbraco CMS

Dynamic Configuration via CMS

- Umbraco CMS controlled every interface aspect: colors, icons, text, routes, calendars, fare rules, promotional logic
- Non-technical users modified business rules in real-time without deployments
- Enabled rapid business rule changes maintaining data integrity and UX consistency

Frontend Excellence

- TypeScript, Angular, JavaScript with component-based design
- Clean separation between smart and dumb components for maintainability and performance
- Scaled features while keeping system lightweight and stable under heavy usage

Foundation Building

- Established foundation in Clean Code and software craftsmanship
- Learned Clean Architecture, Hexagonal Architecture, SOLID principles
- Studied recommended books: Clean Code, Clean Architecture
- Period that shaped engineering mindset for career

Technologies: C#, Umbraco CMS, .NET Framework, Angular, Pattern Design, Dependency Injection, OOP, JavaScript, TFS

DATASOFT | SQL Developer & Database Administrator

 2015 – 2017 |  Manizales, Colombia (On-site)

SaaS platform for Colombian local and regional governments managing public finances: vehicle taxes, traffic fines, debt collection. Platform used by multiple municipal and departmental governments for automated tax liquidation, payment tracking, legal enforcement.

Career Growth

- Started as SQL/PL-SQL Developer, promoted to Database Administrator within 3 months
- Self-taught database administration through books and documentation
- Learned: data normalization, indexing, performance tuning, query optimization, user permissions, access control, table partitioning, backup strategies

Case Study: Query Optimization & Performance Recovery

Challenge: Major monthly process consolidated financial data for Treasury Department reports. Execution time: 2 days for success. Rollback time: 2 days if failed, locking entire database for hours.

Investigation: Performed reverse engineering of database structure. Mapped every table, column, trigger, stored procedure involved. Identified issues: SELECT * queries with unnecessary columns, excessive joins.

Solution: Rewrote queries using only required columns; introduced targeted indexing on frequently accessed fields; created materialized views caching static reference data.

-  Reduced execution time from 2 days to 2 hours (96% improvement)
-  Transformation confirmed promotion to Database Administrator
-  Eliminated database locking issues during business hours

Data Integrity & Disaster Recovery

- Implemented strict data access policies

- Built backup and recovery strategy before DevOps practices were common
- Automated scheduled database backups with manual scripting
- Integrated with Google Drive API for secure off-site storage
- Performed full disaster recovery simulation — deleted production database, restored from backup, validated all data integrity tests successfully

Technologies: PostgreSQL, PL/SQL, SQL, PHP, Excel, Stored Procedures, Triggers, Indexing, Query Optimization, Materialized Views, Google Drive API, Backup & Recovery, Linux (basic), SaaS Architecture

EDUCATION

Systems and Telecommunications Engineering

University of Manizales | Manizales, Colombia | 📅 2016 – 2020

Technology in Computer Systems

Caldas University | Manizales, Colombia | 📅 2012 – 2015

LANGUAGES

Spanish: Native | **English:** B2 (Upper Intermediate / Professional Working Proficiency)

LEADERSHIP & SOFT SKILLS

- Highly motivated professional with proven track record of exceeding expectations
- Effective team player with cross-functional collaboration experience
- Continuous learner: rapidly adapts to new technologies, concepts, and techniques
- Strong communication skills bridging technical and business stakeholders
- Confident, proactive, and organized with excellent time management
- Fast learner: consistently delivers in unfamiliar domains within tight deadlines
- Mentorship: coaches junior developers and establishes best practices
- Client-facing experience: translates business requirements into technical solutions

PROFESSIONAL PHILOSOPHY

I have a passion for implementing good practices and applying clean architectures so that software can be maintained over time by any member of the team. My approach combines technical excellence with business understanding, ensuring that solutions are not only functional but sustainable, scalable, and aligned with organizational goals.

I believe in:

- **Code as Communication:** Writing code that speaks to both machines and humans
- **Architecture as Business Enabler:** Technical decisions driven by business value
- **Quality as Continuous Process:** Not a final step, but an ongoing commitment
- **Knowledge Sharing:** Documentation, mentorship, and team empowerment
- **AI-Augmented Development:** Leveraging AI tools while maintaining human expertise and judgment

INTERESTS & HOBBIES

 Watching Movies  Ping-Pong  Football